Nmap

Nmap is a powerful open-source tool used for **network scanning and security auditing**. It helps cybersecurity professionals discover devices on a network and identify open ports, running services, and potential vulnerabilities.

LAB SETUP

Machines Required:

1. Kali Linux (Attacker)
2. Metasploitable / Target VM
3. Both in same network

Find IP Address
On kali:

```
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.126.128  netmask 255.255.255.0  broadcast 192.168.126.255
        inet6 fe80::f469:a45f:2d71:ee39  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:73:4e:a0  txqueuelen 1000  (Ethernet)
        RX packets 41  bytes 5128 (5.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 38  bytes 4248 (4.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

On Metasploitable/ Target VM:

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:f2:cb:06
          inet addr:192.168.126.129  Bcast:192.168.126.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fef2:cb06/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:41 errors:0 dropped:0 overruns:0 frame:0
          TX packets:68 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4394 (4.2 KB)  TX bytes:6976 (6.8 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:97 errors:0 dropped:0 overruns:0 frame:0
          TX packets:97 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21529 (21.0 KB)  TX bytes:21529 (21.0 KB)

msfadmin@metasploitable:~$ _
```

When you type:

1. **Nmap --version**

It displays the **installed version of Nmap** and related information

```
┌──(kali㉿kali)-[~]
└─$ nmap --version
Nmap version 7.98 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.5.5 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.5 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

2. **Nmap –-help**

It shows a **quick summary of all important options** available in Nmap.

```
┌──(kali㉿kali)-[~]
└─$ nmap --help
Nmap 7.98 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
```

Nmap Scan Control and Host Listing Options

1. -sn

Performs host discovery only (no port scanning).
Used to find which systems are alive in a network.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -sn
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:09 -0500
Nmap scan report for 192.168.126.129
Host is up (0.00096s latency).
MAC Address: 00:0C:29:F2:CB:06 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds
```

2. -sL

Lists the targets without sending any packets.
Useful for verifying the target range before scanning.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -sL
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:09 -0500
Nmap scan report for 192.168.126.129
Nmap done: 1 IP address (0 hosts up) scanned in 0.50 seconds
```

Different Host Discovery Techniques in Nmap

1. -Pn

Skips host discovery and assumes the host is up.
Useful when ICMP/ping is blocked by a firewall.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -Pn
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:09 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0039s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.89 seconds
```

2. -PS

Sends TCP SYN packets to check if hosts are active.
Commonly used to bypass ICMP ping restrictions.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -PS
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:09 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0028s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.83 seconds
```

3. -PA

Sends TCP ACK packets for host discovery.
Helps detect hosts through certain firewall rules.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -PA
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:09 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0023s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.87 seconds
```

4. -PU

Uses UDP packets to identify live hosts.
Helpful when TCP-based discovery is filtered.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -PU
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:10 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0020s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds
```

5. -PY

Uses SCTP INIT packets to detect active systems.
Useful in networks running SCTP services.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -PY
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:10 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0025s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds
```

6. -PE

Sends ICMP Echo request (regular ping).
Basic and most common host discovery method.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -PE
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:10 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0019s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.82 seconds
```

7. -PP

Sends ICMP Timestamp request packets.
Used when echo requests are blocked.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -PP
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:10 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0034s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.82 seconds
```

8. -PM

Sends ICMP Netmask request packets.
Another alternative for ICMP-based discovery.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -PM
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:10 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0020s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.86 seconds
```

9. -PO

Sends IP protocol packets to detect host availability.
Checks response to specific IP protocols instead of ports.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -P0
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:10 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds
```

Comparison:

Option	Full Form	Protocol Used	When to Use	Key Difference
-Pn	No Ping	None	When firewall blocks ping	Skips discovery completely
-PS	TCP SYN Ping	TCP (SYN)	When ICMP is blocked	Sends SYN packet to a port
-PA	TCP ACK Ping	TCP (ACK)	Bypass some firewalls	Sends ACK packet instead of SYN
-PU	UDP Ping	UDP	When TCP is filtered	Sends UDP packet to port
-PY	SCTP INIT Ping	SCTP	Telecom/SCTP networks	Uses SCTP protocol
-PE	ICMP Echo	ICMP	Normal networks	Standard ping request
-PP	ICMP Timestamp	ICMP	When echo is blocked	Sends timestamp request
-PM	ICMP Netmask	ICMP	Rare cases	Sends netmask request
-PO	IP Protocol Ping	IP	Advanced scanning	Uses IP protocol number instead of port

I. TCP Connect Scan (-sT)

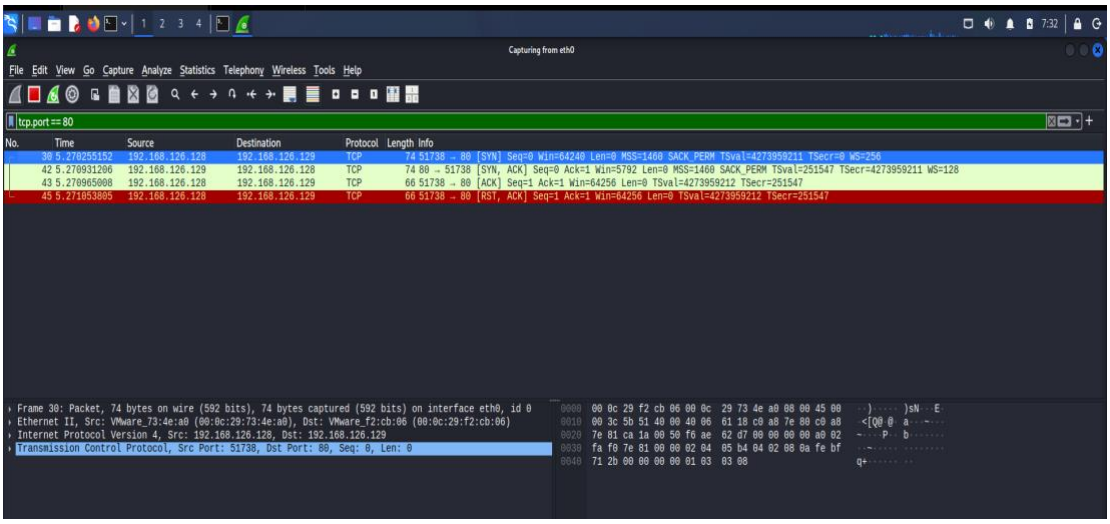
-sT performs a **full TCP three-way handshake** (SYN → SYN-ACK → ACK) to determine if a port is open.
It is used when Nmap does not have raw packet privileges

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -sT
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:31 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds
```

Wireshark Filter:

tcp.port == 80



II. Stealth Scan (-sS)

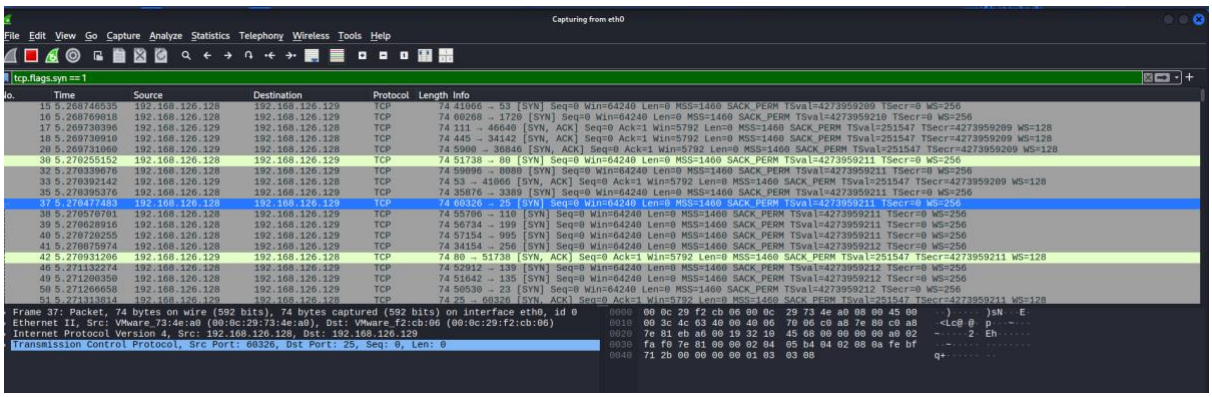
Performs a half-open TCP scan by sending a SYN packet and not completing the handshake.
It is fast and stealthy, commonly used in penetration testing.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -sS
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:40 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0016s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.88 seconds
```

Wireshark Filter:

tcp.flags.syn == 1



III. Inverse TCP Scan (FIN Scan)

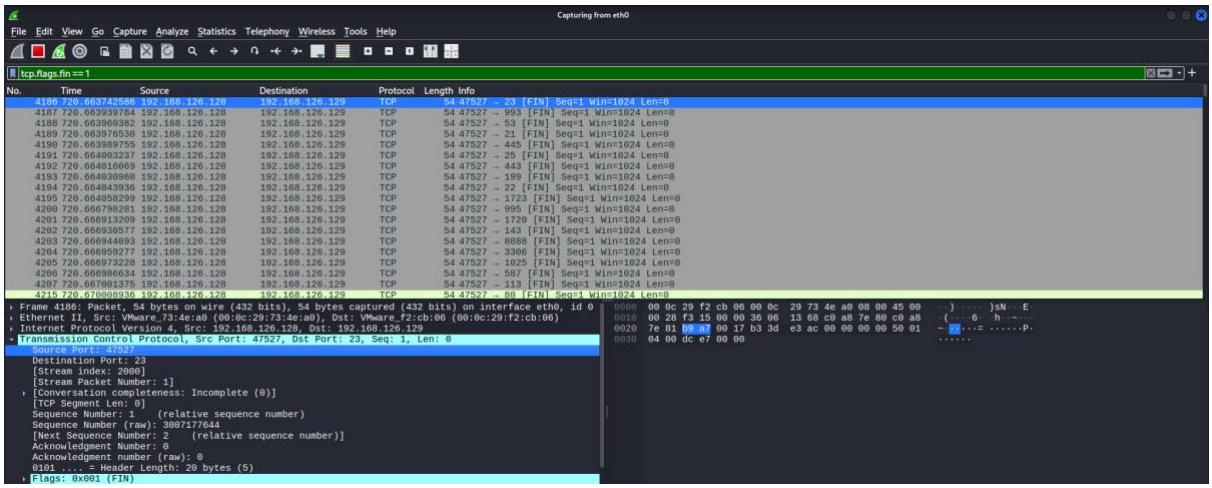
Sends a FIN packet to check port status based on the response.
Useful for bypassing some firewalls and filtering systems.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -sF
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:43 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0012s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE         SERVICE
21/tcp   open|filtered ftp
22/tcp   open|filtered ssh
23/tcp   open|filtered telnet
25/tcp   open|filtered smtp
53/tcp   open|filtered domain
80/tcp   open|filtered http
111/tcp  open|filtered rpcbind
139/tcp  open|filtered netbios-ssn
445/tcp  open|filtered microsoft-ds
512/tcp  open|filtered exec
513/tcp  open|filtered login
514/tcp  open|filtered shell
1099/tcp open|filtered rmiregistry
1524/tcp open|filtered ingreslock
2049/tcp open|filtered nfs
2121/tcp open|filtered ccproxy-ftp
3306/tcp open|filtered mysql
5432/tcp open|filtered postgresql
5900/tcp open|filtered vnc
6000/tcp open|filtered X11
6667/tcp open|filtered irc
8009/tcp open|filtered ajp13
8180/tcp open|filtered unknown
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 2.05 seconds
```

Wireshark Filter:

tcp.flags.fin == 1



IV. ACK Scan

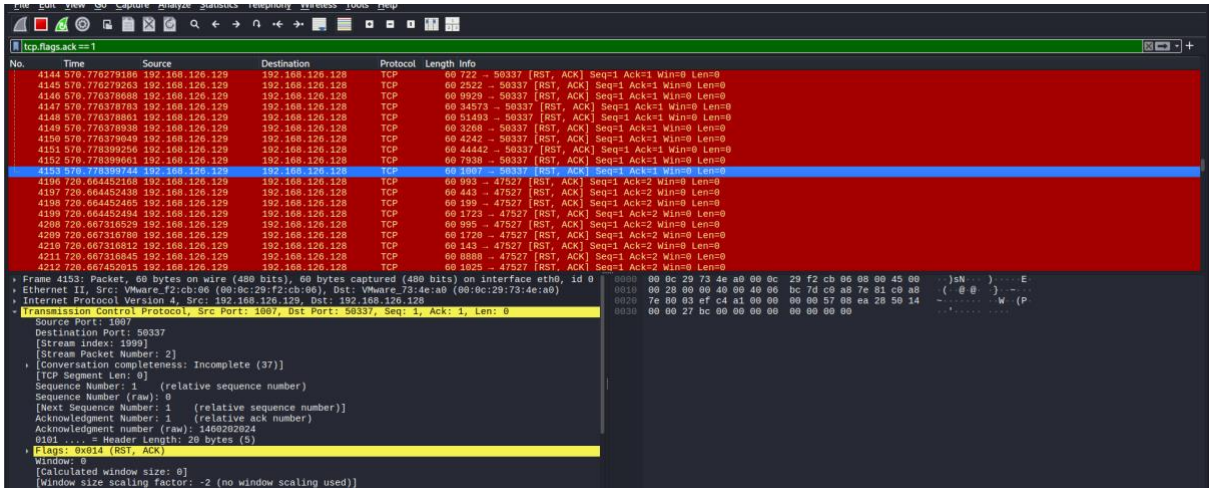
Sends ACK packets to determine firewall rules and filtering.
It does not determine open ports, only whether they are filtered or unfiltered.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.126.129 -sA
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:45 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0025s latency).
All 1000 scanned ports on 192.168.126.129 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.82 seconds
```

Wireshark Filter:

tcp.flags.ack == 1



V. UDP Scan

Scans UDP ports by sending UDP packets to the target.
Used to detect services like DNS, SNMP, and DHCP running on UDP.

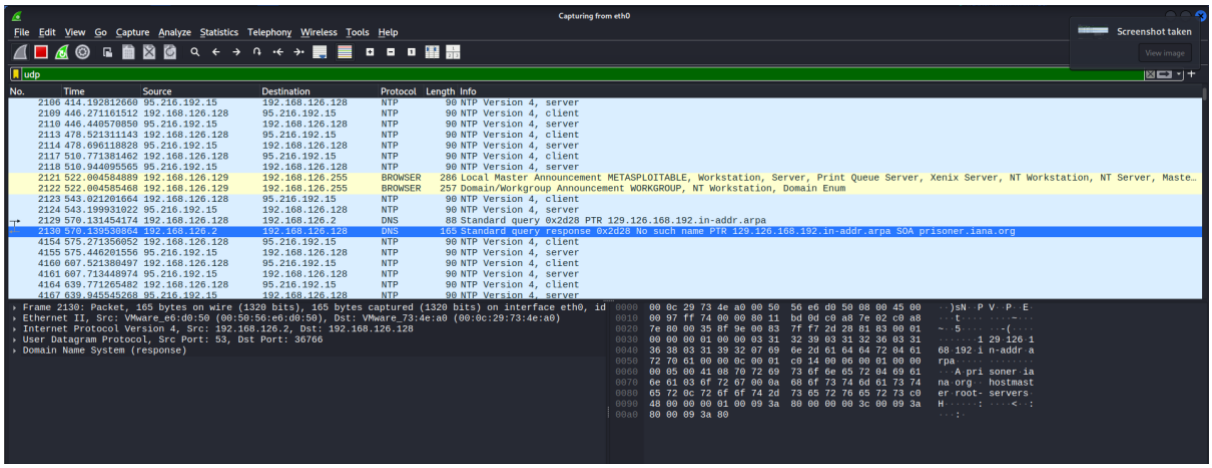
```
┌──(kali㉿kali)-[~]
└─$ nmap -sU -p 53,67,123,161 192.168.126.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 07:57 -0500
Nmap scan report for 192.168.126.129
Host is up (0.0014s latency).

PORT     STATE  SERVICE
53/udp   open   domain
67/udp   closed dhcps
123/udp  closed ntp
161/udp  closed snmp
MAC Address: 00:0C:29:F2:CB:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds
```

Wireshark Filter:

Udp



Analysis of TCP/UDP Scan Traffic Using Wireshark Conversations Statistics

Go to the top menu:

Statistics → Conversations

Inside the Conversations window, choose:

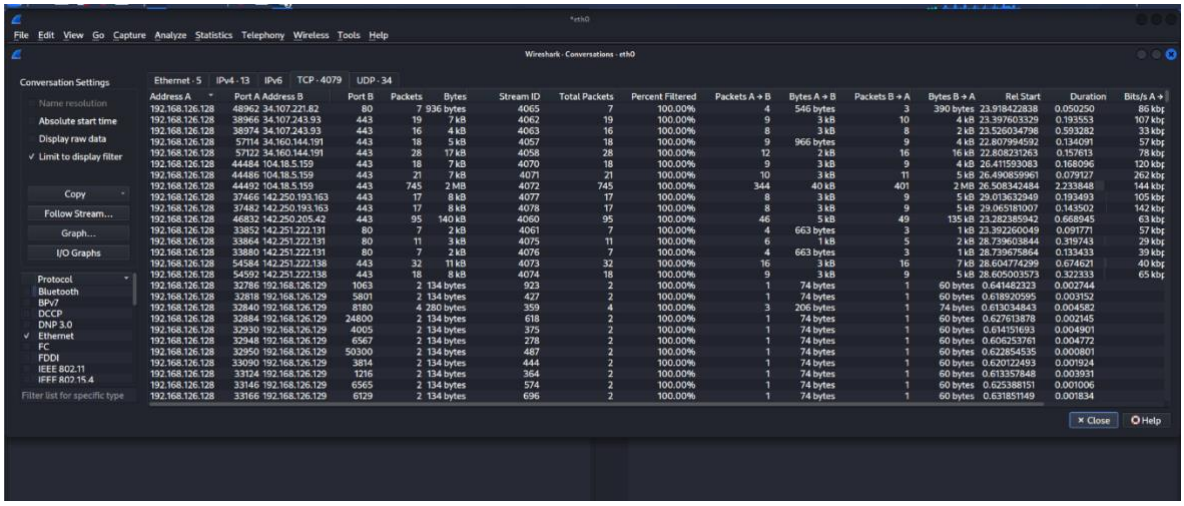
1. **Ethernet** → MAC level
2. **IPv4** → IP level
3. **TCP** → For TCP scan analysis
4. **UDP** → For UDP scan analysis

For Nmap TCP scans → Click **TCP**
For UDP scan → Click **UDP**

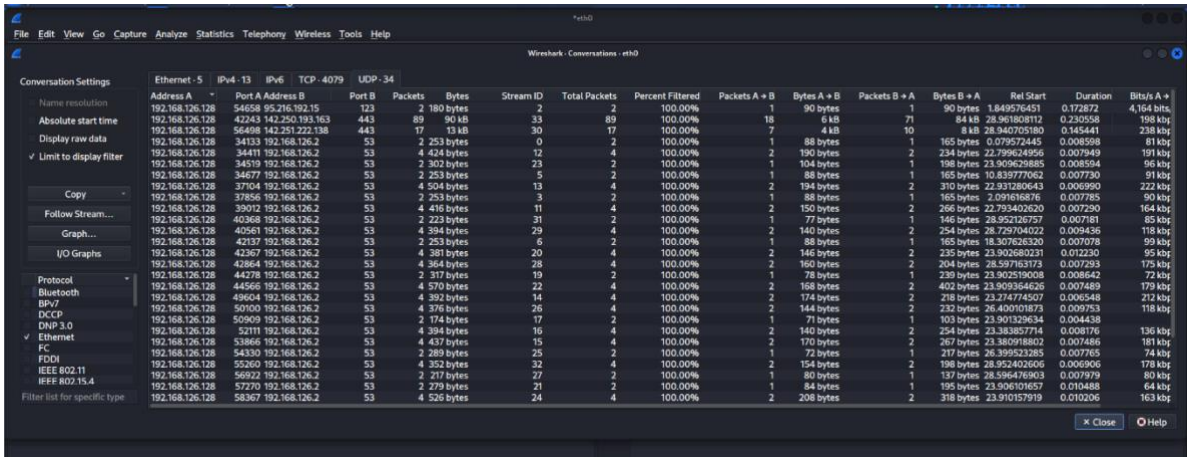
On the left side:

Uncheck: **Limit to display filter**

TCP



UDP



Analysis of Active Network Hosts Using Wireshark Endpoints Statistics

From the top menu go to:

Statistics → Endpoints

Inside the Endpoints window you will see tabs:

- Ethernet → MAC addresses
- IPv4 → IP addresses
- IPv6 → IPv6 addresses
- TCP → TCP endpoints
- UDP → UDP endpoints

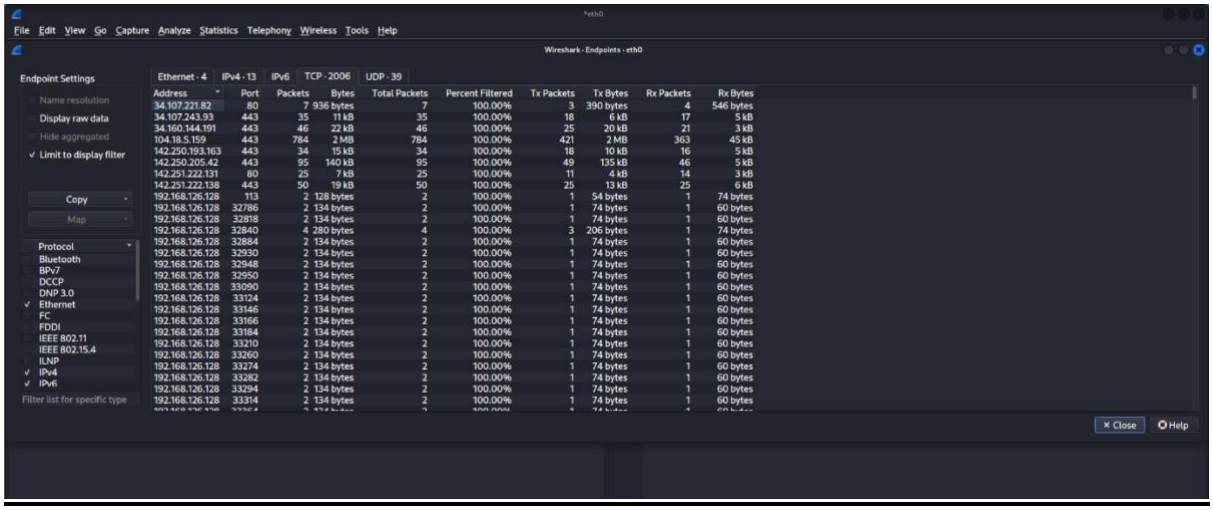
For host discovery analysis → Click **IPv4**
For TCP scan analysis → Click **TCP**
For UDP scan → Click **UDP**

On the left side:

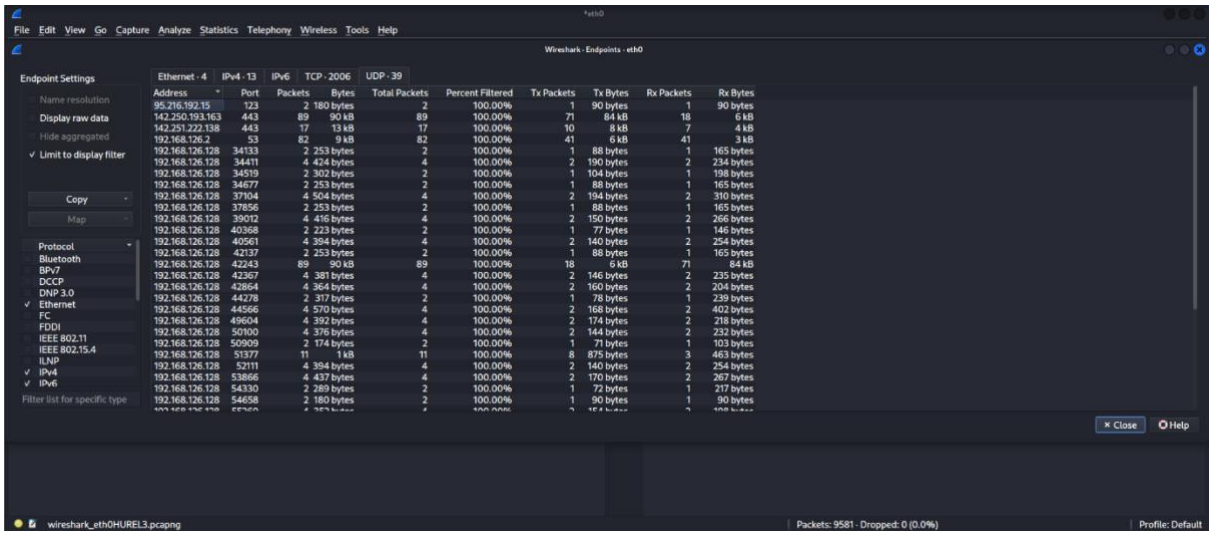
Make sure this is unchecked: **Limit to display filter**

Otherwise, some devices may not appear.

TCP


*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Wireshark · Endpoints · eth0
Endpoint Settings
Name resolution
Display raw data
Hide aggregated
Limit to display filter
Copy
Map
Protocol
Bluetooth
BPv7
DCCP
DNP 3.0
Ethernet
FC
FDDI
IEEE 802.11
IEEE 802.15.4
ILNP
IPv4
IPv6
Filter list for specific type
Ethernet · 4
IPv4 · 13
IPv6
TCP · 2006
UDP · 39
Address Port Packets Bytes Total Packets Percent Filtered Tx Packets Tx Bytes Rx Packets Rx Bytes
34.107.221.82 80 7 936 bytes 7 100.00% 3 390 bytes 4 546 bytes
34.107.243.93 443 35 11 kB 35 100.00% 18 6 kB 17 5 kB
34.160.144.191 443 46 22 kB 46 100.00% 25 20 kB 21 3 kB
104.18.5.159 443 784 2 MB 784 100.00% 421 2 MB 363 45 kB
142.250.193.163 443 34 15 kB 34 100.00% 16 10 kB 16 5 kB
142.250.205.42 443 95 140 kB 95 100.00% 49 135 kB 46 5 kB
142.251.222.131 80 25 7 kB 25 100.00% 11 4 kB 14 3 kB
142.251.222.138 443 50 19 kB 50 100.00% 25 13 kB 25 6 kB
192.168.126.128 113 2 128 bytes 2 100.00% 1 54 bytes 1 74 bytes
192.168.126.128 32786 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 32818 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 32840 4 280 bytes 4 100.00% 3 206 bytes 1 74 bytes
192.168.126.128 32884 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 32930 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 32948 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 32950 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33090 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33124 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33146 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33166 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33184 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33210 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33260 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33274 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33282 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33294 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
192.168.126.128 33314 2 134 bytes 2 100.00% 1 74 bytes 1 60 bytes
Close
Help

UDP


*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Wireshark · Endpoints · eth0
Endpoint Settings
Name resolution
Display raw data
Hide aggregated
Limit to display filter
Copy
Map
Protocol
Bluetooth
BPv7
DCCP
DNP 3.0
Ethernet
FC
FDDI
IEEE 802.11
IEEE 802.15.4
ILNP
IPv4
IPv6
Filter list for specific type
Ethernet · 4
IPv4 · 13
IPv6
TCP · 2006
UDP · 39
Address Port Packets Bytes Total Packets Percent Filtered Tx Packets Tx Bytes Rx Packets Rx Bytes
95.216.192.15 123 2 180 bytes 2 100.00% 1 90 bytes 1 90 bytes
142.250.193.163 443 89 90 kB 89 100.00% 71 84 kB 18 6 kB
142.251.222.138 443 17 13 kB 17 100.00% 10 8 kB 7 4 kB
192.168.126.2 53 82 9 kB 82 100.00% 41 6 kB 41 3 kB
192.168.126.128 34133 2 253 bytes 2 100.00% 1 88 bytes 1 165 bytes
192.168.126.128 34411 4 424 bytes 4 100.00% 2 190 bytes 2 234 bytes
192.168.126.128 34519 2 302 bytes 2 100.00% 1 104 bytes 1 198 bytes
192.168.126.128 34677 2 253 bytes 2 100.00% 1 88 bytes 1 165 bytes
192.168.126.128 37104 4 504 bytes 4 100.00% 2 194 bytes 2 310 bytes
192.168.126.128 37856 2 253 bytes 2 100.00% 1 88 bytes 1 165 bytes
192.168.126.128 39012 4 416 bytes 4 100.00% 2 150 bytes 2 266 bytes
192.168.126.128 40368 2 223 bytes 2 100.00% 1 77 bytes 1 146 bytes
192.168.126.128 40561 4 394 bytes 4 100.00% 2 140 bytes 2 254 bytes
192.168.126.128 42137 2 253 bytes 2 100.00% 1 88 bytes 1 165 bytes
192.168.126.128 42243 89 90 kB 89 100.00% 18 6 kB 71 84 kB
192.168.126.128 42367 4 381 bytes 4 100.00% 2 146 bytes 2 235 bytes
192.168.126.128 42864 4 364 bytes 4 100.00% 2 160 bytes 2 204 bytes
192.168.126.128 44278 2 317 bytes 2 100.00% 1 78 bytes 1 239 bytes
192.168.126.128 44566 4 570 bytes 4 100.00% 2 168 bytes 2 402 bytes
192.168.126.128 49604 4 392 bytes 4 100.00% 2 174 bytes 2 218 bytes
192.168.126.128 50100 4 376 bytes 4 100.00% 2 144 bytes 2 232 bytes
192.168.126.128 50909 2 174 bytes 2 100.00% 1 71 bytes 1 103 bytes
192.168.126.128 51377 11 1 kB 11 100.00% 8 875 bytes 3 463 bytes
192.168.126.128 52111 4 394 bytes 4 100.00% 2 140 bytes 2 254 bytes
192.168.126.128 53866 4 437 bytes 4 100.00% 2 170 bytes 2 267 bytes
192.168.126.128 54330 2 289 bytes 2 100.00% 1 72 bytes 1 217 bytes
192.168.126.128 54658 2 180 bytes 2 100.00% 1 90 bytes 1 90 bytes
Close
Help
wireshark_eth0HUREL3.pcapng
Packets: 9581 · Dropped: 0 (0.0%)
Profile: Default